already shown that chemical forces may also be regarded in general as exchange forces. For a similar exchange of charges occurs in the case of chemical forces, too. The simplest example of this is the hydrogen molecule ion, which consists of a hydrogen atom and a hydrogen nucleus (Figure 17). It is therefore actually a structure composed of two protons with an electron circling around them. This ion is a truly stable structure, and the force which holds it together owes its existence to the circumstance that the one electron revolves at times around one proton, at times around the other one. This means that in this case, too, we find that the force is linked with an exchange of charge—the shift of the electron from one proton to the other.

The concept of ‘exchange force’ can be comprehended most easily on the ground of the following experiment, carried out with the big Lawrence cyclotron in California in 1948 (cf. page 155). Neutrons of great energy are hurled against protons in a cloud chamber. The paths of the protons after collision become visible in the cloud chamber. In the case of an ordinary force, one would expect the great majority of the neutrons to be deflected to a very slight extent only (since they would not hit the proton exactly in the centre), while the protons would be hurled aside, with a relatively low velocity, at an angle of 90 deg. to the path of the neutrons. But in the case of an exchange force, the neutrons and protons must exchange roles after the collision; in the majority of collisions, the protons must continue along the paths of the oncoming neutrons (since the neutrons have actually been changed into protons) while the neutron is hurled aside at an angle of approximately 90 deg. This is exactly what we actually see in the cloud chamber. Figure 17A shows the tracks of the protons, most of which fly on along an almost straight path in the direction in which the oncoming neutrons were travelling. A magnetic field causes their paths to become more or less strongly curved, according to their energy. (The straight lines slanted at less than 45 deg. are produced by a thin wire mesh in the cloud chamber.)

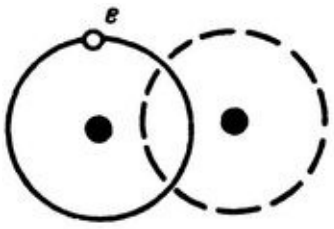


Figure 17.—Ion of hydrogen molecule.